Exact Intertwining and Uniform Clustering in the Reconstructed Site Form for Coupled $\mathbb{Z}_2$ Slices

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Abstract

For a finite reflection-positive coupled $\mathbb{Z}_2$ slice, we identify the connected two-point function written directly with the reconstructed site form and the forced transfer operator with the standard connected correlator of a symmetric Euclidean matrix operator. The identification is exact at every time separation. It follows that the already established Dobrushin fluctuation bound gives

$$|C_L^{\mathrm{rec}}(u;n)| \le \|u\|_2^2 e^{-mn}$$

with one $m > 0$ chosen before the spatial extent L. The proof preserves both the exponent and the prefactor. Nine interface theorems are elaborated by Lean 4 against Mathlib, including the finite-iterate intertwining, the exact correlator identity, and the reconstructed-site clustering endpoint. The new contribution is this operator/correlator composition. The small-coupling Dobrushin estimate is inherited unchanged; no new analytic estimate or continuum reconstruction is claimed.

1 Result and claim boundary

Reflection positivity supplies a positive form on boundary data and makes a time-translation operator meaningful in the reconstructed space [1, 2, 3]. A spectral estimate written for an auxiliary Euclidean matrix, however, is not yet a bound on the correlator expressed in that physical form. One must identify the two coordinate descriptions, transport all finite powers, and check that the vacuum subtraction agrees. This paper carries out exactly those steps for the coupled $\mathbb{Z}_2$ slice.

The endpoint is stronger than an operator-only interface: the reconstructed connected correlator itself obeys a volume-uniform exponential estimate. It is also deliberately narrower than a general Osterwalder–Schrader (OS) reconstruction theorem. We do not prove a new contraction estimate, a thermodynamic or continuum limit, or a Wightman reconstruction. The analytic input is a previously checked finite-state Dobrushin theorem. The present work proves that its gap controls the correlator defined by the previously checked reflection-positive site form, without a change of constants.

2 Three descriptions of the same finite slice

For $L \in \mathbb{N}$, let

$$\Sigma_L = (\mathrm{Fin}\, L \to \mathrm{Fin}\, 2)$$

be the finite set of spin configurations. Fix a strictly positive weight $w : \Sigma_L \to \mathbb{R}_{>0}$ and the real symmetric spatial kernel $K_\beta(\sigma, \tau)$. The reconstructed site form on complex boundary vectors is

$$\langle f, g\rangle_{\mathrm{site},w} = \sum_{\sigma\in\Sigma_L} \overline{f(\sigma)}g(\sigma)w(\sigma)^{-1}. \tag{1}$$

The transfer operator forced by the site–bond identity is

$$(T_{w,\beta}g)(\sigma) = w(\sigma) \sum_{\tau\in\Sigma_L} K_\beta(\sigma, \tau)g(\tau). \tag{2}$$

The symmetric weighted matrix used in the spectral lane is

$$S_{w,\beta}(\sigma, \tau) = \sqrt{w(\sigma)}K_\beta(\sigma, \tau)\sqrt{w(\tau)}. \tag{3}$$

These objects were already present in the formal development. The bridge is the square-root dressing

$$(Q_wu)(\sigma) = \sqrt{w(\sigma)}u(\sigma), \qquad u : \Sigma_L \to \mathbb{R}, \tag{4}$$

with the result regarded as a complex boundary vector.

Proposition 2.1 (Isometry and one-step intertwining)**.** *If $w(\sigma) > 0$ for every σ, then for real u, v,*

$$\langle Q_wu, Q_wv\rangle_{\mathrm{site},w} = \sum_\sigma u(\sigma)v(\sigma), \tag{5}$$

$$T_{w,\beta}(Q_wu) = Q_w(S_{w,\beta}u). \tag{6}$$

Consequently, for every $n \in \mathbb{N}$,

$$T^{;n}_{w,\beta}(Q_wu) = Q_w(S^{;n}_{w,\beta}u). \tag{7}$$

The equality (5) cancels $\sqrt{w(\sigma)}^2/w(\sigma)$ pointwise. Equation (6) is the exact conjugation of the forced transfer operator, not a comparison in norm. Equation (7) follows by induction, so no diagonalisation or unrecorded spectral assumption enters.

Let $\lambda \in \mathbb{R}$ and set

$$P_{w,\beta,\lambda}(\sigma, \tau) = \lambda^{-1}S_{w,\beta}(\sigma, \tau). \tag{8}$$

Scalar multiplication on the complex boundary space and matrix action on the real Euclidean space obey

$$(\lambda^{-1}T_{w,\beta})^n(Q_wu) = Q_w(P^{;n}_{w,\beta,\lambda}u) \qquad (n \in \mathbb{N}). \tag{9}$$

The formal statement uses function iteration for the bare matrix action; a separate lemma identifies it exactly with powers of the continuous linear operator opOf(P) on finite Euclidean space.

3 The reconstructed connected correlator

Let $\Omega : \Sigma_L \to \mathbb{R}_{>0}$ be a Perron vector and write

$$\widehat{\Omega}(\sigma) = \frac{\Omega(\sigma)}{\sqrt{\sum_\tau \Omega(\tau)^2}}. \tag{10}$$

For a real boundary observable u and a time separation n, define directly in the reconstructed site coordinates

$$\begin{aligned} C^{\mathrm{rec}}_{w,\beta,\lambda,\Omega}(u;n) :=\ & \operatorname{Re}\left\langle Q_w u, (\lambda^{-1}T_{w,\beta})^n Q_w u\right\rangle_{\mathrm{site},w} \\ & - \operatorname{Re}\left\langle Q_w\widehat{\Omega}, Q_w u\right\rangle^{2}_{\mathrm{site},w}. \end{aligned} \tag{11}$$

This is the connected, or truncated, two-point function: the second term is the squared vacuum expectation. Both terms use the same reconstructed form and the same dressing map.

The spectral lane uses the real Euclidean correlator

$$C_P(u;n) = \langle u, (\mathrm{opOf}\, P)^n u\rangle_2 - \langle \widehat{\Omega}, u\rangle_2^2. \tag{12}$$

The central new identity closes the gap between (11) and (12).

Theorem 3.1 (Exact correlator identification)**.** *For every positive w, every real β, λ, every real Ω, u, and every $n \in \mathbb{N}$,*

$$C^{\mathrm{rec}}_{w,\beta,\lambda,\Omega}(u;n) = C_{P_{w,\beta,\lambda}}(u;n). \tag{13}$$

The proof applies (9) to the time-dependent term, (5) to both pairings, and the matrix-power lemma to change from iteration to $\mathrm{opOf}(P)^n$. It is an equality for every finite n; there is no limiting argument and no loss of exponent or prefactor.

4 Gap implies reconstructed-site clustering

Let

$$\Pi_\Omega v = \langle \widehat{\Omega}, v\rangle_2 \widehat{\Omega}, \qquad P_\perp = \mathrm{opOf}(P) - \Pi_\Omega.$$

If P is symmetric, $P\Omega = \Omega$, and Ω is positive, the formal library packages $\mathrm{opOf}(P)$ and $\widehat{\Omega}$ as a vacuum transfer pair. The existing gap-to-clustering theorem gives

$$|C_P(u;n)| \le \|u\|_2^2 r^n \quad \text{whenever} \quad \|P_\perp\| \le r. \tag{14}$$

Combining this with Theorem 3.1 yields the physical-form endpoint.

Corollary 4.1 (Reconstructed-site decay)**.** *Under the hypotheses above, if $\|P_\perp\| \le r$, then*

$$|C^{\mathrm{rec}}_{w,\beta,\lambda,\Omega}(u;n)| \le \|u\|_2^2 r^n \tag{15}$$

for every real boundary vector u and every $n \in \mathbb{N}$.

The result preserves the exact constant in (14). In particular, it does not replace r by a larger comparison rate or introduce a volume-dependent equivalence-of-norms factor.

5 The volume-uniform endpoint

For the coupled-slice weight $w = \mathrm{sliceW}(\gamma, L)$, the inherited Dobrushin theorem supplies a positive exponent before the spatial size is chosen. The formal convention uses configurations on Σ_{L+1}; we keep that binder rather than silently renumbering it.

Theorem 5.1 (Uniform reconstructed-site clustering)**.** *Let $0 < \alpha < 1$ and suppose*

$$2 \tanh |\beta| + 2 \tanh |\gamma| \le \alpha. \tag{16}$$

Then there exists $m > 0$ such that, for every $L \in \mathbb{N}$, there are a positive Perron value λ_L and a positive vector $\Omega_L : \Sigma_{L+1} \to \mathbb{R}_{>0}$ satisfying

$$P_L \Omega_L = \Omega_L, \tag{17}$$

$$\|\mathrm{opOf}(P_L) - \Pi_{\Omega_L}\| \le e^{-m}, \tag{18}$$

where $P_L = P_{\mathrm{sliceW}(\gamma,L),\beta,\lambda_L}$. Moreover, for all real $u : \Sigma_{L+1} \to \mathbb{R}$ and all $n \in \mathbb{N}$,

$$\left| C^{\mathrm{rec}}_{\mathrm{sliceW}(\gamma,L),\beta,\lambda_L,\Omega_L}(u; n) \right| \le \|u\|_2^2 e^{-mn}, \tag{19}$$

and the exact intertwining (9) *holds.*

The quantifier order

$$\exists m > 0 \; \forall L \; \exists (\lambda_L, \Omega_L)$$

is part of the theorem type. This, rather than the mere existence of a finite-volume Perron gap for each L, is the asserted uniformity. The new clause (19) makes the endpoint a statement about the reconstructed connected correlator, not only an auxiliary operator norm.

6 Dependency ledger: inherited and proved here

The proof has a short exact layer on top of a longer analytic dependency. Keeping them separate prevents a compositional result from being advertised as a new estimate.

Layer	Status in this paper	Mathematical role
Reflection-positive site form and forced $T_{w,\beta}$	inherited	identifies the reconstructed finite boundary space and its time translation
Symmetric weighted kernel and Perron tilt	inherited	supplies the real Euclidean operator used by the spectral lane
Dobrushin window (16) and bound (18)	inherited unchanged	chooses m before L
Square-root isometry and all-iterate intertwining	proved in the interface module	identifies reconstructed and Euclidean coordinates exactly
Matrix powers versus bare action iteration	proved in the interface module	aligns the two implementations of finite time evolution
Identity (13) and decay (15)	proved in the interface module	transports the gap to the reconstructed connected correlator without loss
Uniform endpoint (19)	proved by composition	exposes the physical-form conclusion with the correct binder order

This ledger also describes the paper's autonomy. The result depends on machine-checked repository modules, but its new claim is not merely that those files coexist: the exact correlator equality and its uniform corollary are new theorem declarations whose types mention both lanes.

7 Formal artifact and reproducibility

The authoritative file is `YangMills/OS/OSReconstructionUniform.lean`. It contains two definitions and nine theorem declarations:

Lean declaration	Role
`siteForm_qEmbed`	reconstructed-form isometry
`transferOp_qEmbed`	one-step forced-operator intertwining
`transferOp_iterate_qEmbed`	unnormalised finite iterates
`transferOp_qEmbed_tilt`	one-step Perron-normalised intertwining
`transferOp_qEmbed_tilt_iterate`	normalised finite iterates
`opOf_pow_toLp_act`	operator powers equal matrix-action iteration
`reconstructedConnCorr_eq_connCorr`	exact identity (13)
`reconstructedConnCorr_decay`	reconstructed-site decay (15)
`os_reconstruction_uniform_gap`	uniform theorem, including (19)

Validation used a Colab Pro+ CPU/high-RAM runtime with the GPU disabled; no Lean or Lake command ran on the local Windows desktop. The target build completed all 8191 jobs with exit code zero. The exact LF source had 8682 bytes and SHA-256

`199229725d2d359e19a08792317d7f7e552cf027990505a4a4ffb2542295437a`,

while the materialised `.olean` had SHA-256

`2741f4c24178f2ea632d914f309acc71c714b831a5c5b282b9e8bd8a94a34478`.

A separate verification manifest records timestamps, toolchain pins, oracle counters, byte counts, and attacks attempted. Compilation and counters are evidence for an independent audit, not a self-issued terminal verdict.

8 Relation to primary literature

The OS papers formulate Euclidean reconstruction and reflection positivity [1, 2]; Osterwalder and Seiler develop the lattice-gauge setting [3]; and Fröhlich, Israel, Lieb, and Simon use reflection positivity in lattice systems [4]. Dobrushin's conditional-probability criterion is the primary source for the high-temperature uniqueness mechanism [5]. These are conceptual and analytic antecedents, not claims of novelty.

The closest public papers in the author's own sequence separate the reflection-positive site form, the transfer bridge, the finite-volume Perron vacuum, and fluctuation-sector bounds. The present paper's distinct endpoint is the exact equality (13) followed by the reconstructed-site bound (19). We make no global priority claim: unindexed formal developments may exist, and bibliographic search is not a mathematical proof of precedence.

9 Limitations

The state space is finite and specific to coupled $\mathbb{Z}_2$ slices. The paper does not prove a general OS theorem, a continuum field theory, or a Yang–Mills mass gap. It does not improve the Dobrushin region (16) or compute a sharper m. The observable is a real boundary vector; extensions to larger observable algebras are not addressed. Finally, (19) is an exponential time-separation estimate inside the reconstructed finite-volume site space. Calling it more than that would exceed the formal theorem.

10 Conclusion

The square-root dressing does more than match two transfer matrices: it matches their iterates, vacuum expectation, and connected correlator. That identity turns the inherited uniform fluctuation bound into an explicit reconstructed-site clustering theorem with the same rate and prefactor. The result is a compact but nontrivial endpoint whose dependency, quantifier order, and limitations are all visible in the machine-checked statement.

References

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